

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Commercial LLM API providers may covertly substitute cheaper models. This work shows software-based detection is unreliable and proposes TEEs as a provably secure alternative.

Based on arxiv.org/abs/2504.04715

ML Researchers • AI Security
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The Model Substitution Problem

Economic Incentives Drive Covert Swaps

The Core Problem

Commercial LLM API providers charge for specific models (e.g., Llama-3-70B) but the black-box nature of APIs means users have no guarantee the advertised model is actually being served. Providers face strong economic incentives to covertly substitute cheaper alternatives.

Four Substitution Types

- 1 Smaller models (e.g., 8B instead of 70B)
- 2 Quantized variants (FP8/INT8 instead of FP16)
- 3 Fine-tuned or updated versions
- 4 Entirely different model families

Formal Hypothesis Testing Framework

H_0 (Null): API returns samples from specified model distribution $P_{\text{spec}}(y|x)$

H_1 (Alternative): API returns samples from actual model distribution $P_{\text{actual}}(y|x)$

Can an auditor distinguish P_{spec} from P_{actual} given only API samples?

FP8 Quantization Preserves Most Accuracy

Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

Key Insight: For most models, FP8 scores fall within error bars of original — detection is extremely challenging

Higher Temperature Amplifies Performance Gaps

But Also Increases Noise

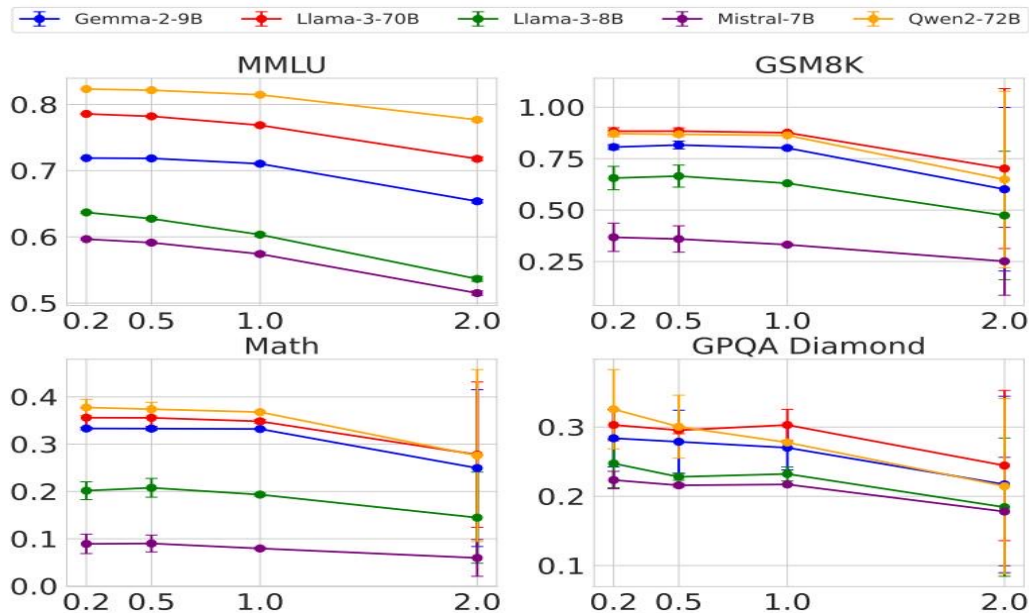


Figure 1 Analysis

At temperature 2.0, performance degrades across all models

The gap between original and substitute narrows under high temperature

High temperature introduces statistical noise that complicates detection

Four-panel chart shows accuracy vs temperature for all five model families

Implication: Temperature manipulation is a simple countermeasure against statistical detection methods

Log Probability Fingerprinting Fails

Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The Nondeterminism Problem

The same Llama-3-8B model produces different log probabilities depending on:

- Software: transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- Hardware: A100 vs H100 GPU differences
- Production-level inference variations

An auditor cannot distinguish between a legitimate model on different infrastructure and an actual substitution.

Randomized Substitution at Low Rates

Makes Statistical Detection Query-Prohibitive

The Randomized Attack

Provider mixes substitute responses with probability p :

$$P_{\text{mixed}} = (1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{substitute}}$$

As $p \rightarrow 0$, the mixed distribution converges to P_{spec} , requiring exponentially more queries to detect the substitution.

Detection Complexity Explodes

$p = 0.5 \rightarrow \sim 10^3$ queries

$p = 0.1 \rightarrow \sim 10^5$ queries

$p = 0.01 \rightarrow \sim 10^7$ queries

Minimum queries for 99% detection power

Additional Adversarial Countermeasure: Benchmark Evasion

Providers can detect known benchmark prompts and route them to the genuine model while serving substitutes for regular traffic — making all software-only audits fundamentally unreliable.

Software-Only Auditing Methods

Are Fundamentally Unreliable

Method Type	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g., MMD)	Require impractically large query budgets; fail against subtle or randomized substitutions	Exponentially more queries needed as substitution probability decreases
Log-Probability Fingerprinting	Defeated by production-level inference nondeterminism across software versions and GPU hardware	Cannot distinguish legitimate infrastructure differences from actual model substitution
Adversarial Countermeasures	Benchmark evasion and randomized routing further undermine all detection approaches	Makes software-only detection fundamentally query-prohibitive

Conclusion: No software-based method can reliably detect model substitution in production LLM APIs under realistic adversarial conditions.

Trusted Execution Environments

Provide Provable Cryptographic Model Integrity

How TEEs Provide Guarantees

- Hardware-secured enclaves with cryptographic attestation
- Prove specific model weights are loaded
- Verify correct inference code is running
- Only modest performance overhead

TEE Properties vs. Software Methods

Provable	S: Not provable	TEE: Cryptographically verifiable
Efficient	S: High query cost	TEE: Modest overhead
Robust	S: Defeated by countermeasures	TEE: Immune to evasion

Practical Deployment Path: NVIDIA Confidential Computing on H100 GPUs

Unlike ZKPs which are computationally prohibitive for large models, TEEs are practical and deployable today.

Key Takeaways

Close the Verification Gap with Hardware Attestation

- 1 Model substitution is economically motivated and practically undetectable via software
- 2 FP8 quantization barely affects benchmarks — making it an attractive but nearly undetectable substitution
- 3 Inference nondeterminism kills log-probability fingerprinting in production environments
- 4 Randomized substitution defeats statistical tests, requiring exponentially more queries
- 5 TEEs are the only currently viable path to verifiable LLM APIs

Call to Action: The research community must prioritize hardware attestation standards for AI inference verification.